

Review

Thermal Management Systems for Lithium-Ion Batteries for Electric Vehicles: A Review

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Abstract

Recently, electric vehicles (EVs) have proven to be a practical option for lowering greenhouse gas emissions and reducing reliance on fossil fuels. Lithium-ion batteries, at the core of this innovation, require efficient thermal management to ensure optimal performance, safety, and durability. This article reviews current scientific studies on controlling the temperature of lithium-ion batteries used in electric vehicles. Several cooling strategies are discussed, including air cooling, liquid cooling, the use of phase change materials (PCMs), and hybrids that combine these three types of cooling, with the primary objective of enhancing the thermal performance of the batteries. Additionally, the challenges and proposed solutions in battery pack design and energy management methodologies are explored. As the demand for electric vehicles increases, improving battery thermal management systems (BTMSs) is becoming increasingly important. Implementing and developing better BTMSs will help increase the autonomy and safety of electric vehicles in the long term.

Keywords: lithium-ion batteries; electric vehicle; battery thermal management system; battery pack design; battery management system



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1. Introduction

In recent years, electric vehicles have established themselves as an excellent option for reducing greenhouse gases and dependence on fossil fuels. Lithium-ion batteries are the main element in electric vehicles. However, a disadvantage of this type of battery is that it requires proper thermal management to ensure its optimal performance, durability, and safety. Figure 1 illustrates some of the causes of a thermal runaway and its physical characteristics that may occur during a short circuit, such as smoke, fire, or even an explosion.

Thermal management systems (BTMSs) are essential to keep the battery pack within a suitable temperature range. Correct thermal management prevents premature aging of the battery pack. Since batteries are sensitive to temperature variations, high temperatures generate harmful chemical reactions, and low temperatures can reduce the efficiency and capacity of the battery pack.

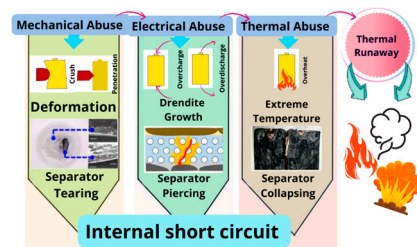


Figure 1. Causes of thermal runaway.

As the demand for electric vehicles grows, research and development of efficient thermal management systems becomes increasingly essential. Implementing advanced BTMSs will improve EVs' range and performance, ensuring the safety and long-term sustainability of this transformative technology.

This article reviews several recent scientific studies that address various techniques and approaches for the thermal management of lithium-ion batteries in electric vehicles. Researchers have explored multiple strategies for optimizing battery thermal performance, including air and liquid cooling, the use of phase change materials (PCMs), and hybrid systems. Additionally, this review examines key challenges in battery pack design and thermal management for lithium-ion batteries in EVs, emphasizing material selection and energy optimization. It explores air, liquid, PCM, and hybrid cooling strategies, addressing technical limitations and innovative solutions to enhance thermal efficiency and stability. Additionally, it provides a structured analysis of battery pack configurations to minimize thermal impact, guiding the selection of the most suitable system for improved safety and efficiency in electric mobility.

Furthermore, the review discusses advancements in energy management methodologies, including smart thermal regulation techniques and multi-objective optimization approaches that balance performance, longevity, and cost-effectiveness. It highlights the role of GIS-based urban planning in strategically placing charging stations to improve accessibility and demand forecasting. By integrating findings from previous studies on genetic algorithms for system optimization, this research is a foundation for developing next-generation thermal management solutions that contribute to electric vehicle sustainability and widespread adoption.

The article has three main parts: introduction, development, and conclusions. The introduction presents a description of the current problems and challenges, as well as the study's objective and structure. The developmental methodology and decision-making process for including articles as state-of-the-art are also discussed. Additionally, graphs and tables are included to classify each article and describe the authors' approaches. Finally, the conclusions provide a starting point for future work on electric vehicle battery pack problems and significant challenges.

2. Methodology

This article presents a review of the state of the art, with the primary purpose of identifying current trends in research and pinpointing areas that require further attention, which is crucial for guiding future research. In addition, a summary of the articles can serve as a quick reference for researchers and practitioners, saving them time when searching for specific information. The search was conducted using the following databases and search engines: IEEEExplore, Springer Nature, ScienceDirect, and SciELO. The research methodology followed to obtain the articles for this state-of-the-art review is illustrated in the diagram in Figure 2, which shows how decisions were made to aggregate the articles.

Figure 3 illustrates the publishing editorials as of 2018 in the five areas listed in Table 1: liquid cooling, air cooling, phase change materials, hybrids, and articles covering various types of cooling.

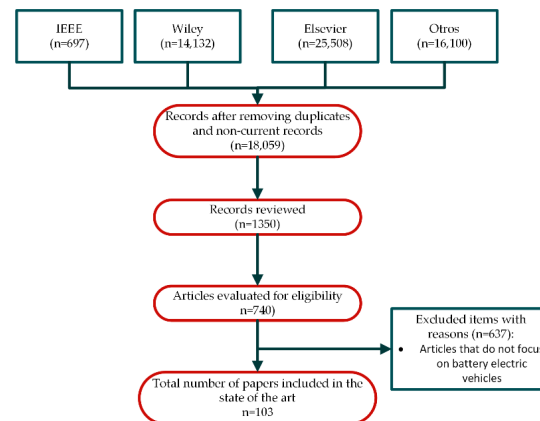


Figure 2. Research methodology to obtain the state of the art.

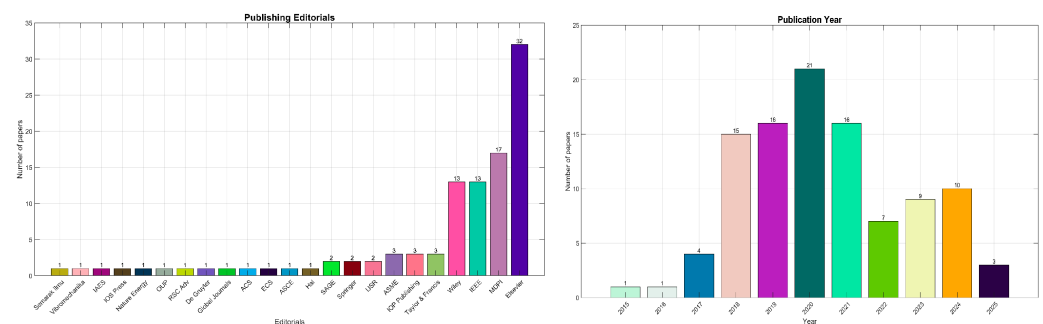


Figure 3. Editorials and years of publication on battery electric vehicles.

Table 1. Approach of thermal management systems.

Approach	References	Summary
Liquid Cooling	[1–8]	In some articles, it can be seen that they perform optimizations of the water-cooling structure, as well as simulations of the cooling characteristics of an active thermal management system or numerical analyses of cooling plates with different structures, among other aspects.
Air Cooling	[9–16]	In some articles, it can be seen that they perform optimizations of air thermal management systems, structural optimizations, or spacing of a battery pack, as well as improvements in cooling performance and design of the structure in parallel thermal management systems, among other aspects.
Phase Change Materials	[17–26]	In some articles, it can be seen that they introduce atomization cooling, analyze the use of PCM as expanded graphite to enhance heat transfer, investigate the thermal properties of PCM, improve thermal performance by utilizing PCM, and modify the geometry of the fins, among other aspects.
Hybrids	[27–30]	In some articles, it is evident that a hybrid thermal management system is being developed, incorporating PCM and metal foam, as well as thermoelectric cooling, among other aspects.
Multiple Types of Cooling	[31–50]	In most articles in this section, summaries of the four primary cooling methods are presented.

The most recent studies on electric vehicles, published between 2018 and 2025, as shown in Figure 3, have become highly relevant due to technological advances and increased adoption of electric mobility. These studies focus on key aspects such as battery thermal management system optimization and improved autonomy, safety, and sustainability of these vehicles. They also present innovative solutions to address challenges such as resource availability, energy efficiency, and the development of charging infrastructure. During this period, research has played an essential role in transitioning to more sustainable and efficient transportation, reducing carbon emissions, and fulfilling global environmental objectives.

3. Results

The literature on battery thermal management (BTMS) and battery pack design can be classified into three main approaches. First, studies devoted to BTMS design address cooling strategies, material selection, and thermal control to improve stability and energy efficiency. Second, research on battery pack configuration analyzes cell layout and structural integration to optimize performance and reduce thermal impact. Finally, papers exploring optimization techniques implement advanced computational models and algorithms to improve thermal management and system efficiency. This classification facilitates the comparison of methodologies and highlights key trends in the development of electric vehicle batteries.

3.1. Areas of Publication

The number of publications analyzed and classified is 99 articles. These articles were classified into four main areas, including electric vehicle design, thermal management system, battery pack design and challenges, and others, as shown in Figure 4.

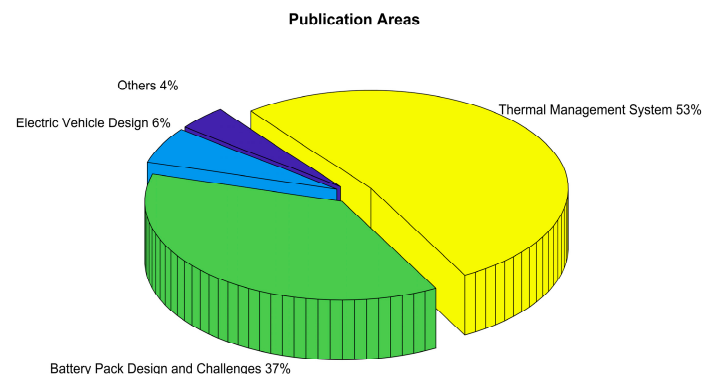


Figure 4. Percentage of publications in the four areas of classification.

Figure 5 shows a diagram that provides an overview of the subareas that comprise each of the main areas mentioned in Figure 4. The various categories include requirements and challenges, modeling and simulation, and the types of cooling used in battery pack thermal management systems.

3.2. Battery Thermal Management System (BTMS)

Battery thermal management systems (BTMSs) are essential in electric vehicles (EVs), as they ensure the safe and stable performance of lithium-ion batteries (LiBs). Their primary function is to regulate temperature, prevent failures, and extend battery life while maintaining efficiency and safety in both mobility and energy storage.

A summary of some of these areas and the authors' focus is presented. The area with the highest percentage is BTMSs, which are essential for maintaining batteries within an optimal temperature range, avoiding degradation, and improving safety and battery

lifetime. The studies reviewed cover various thermal management techniques, including air-cooled, liquid, phase change materials (PCMs), and hybrid systems. Table 1 shows the approach to thermal management systems.

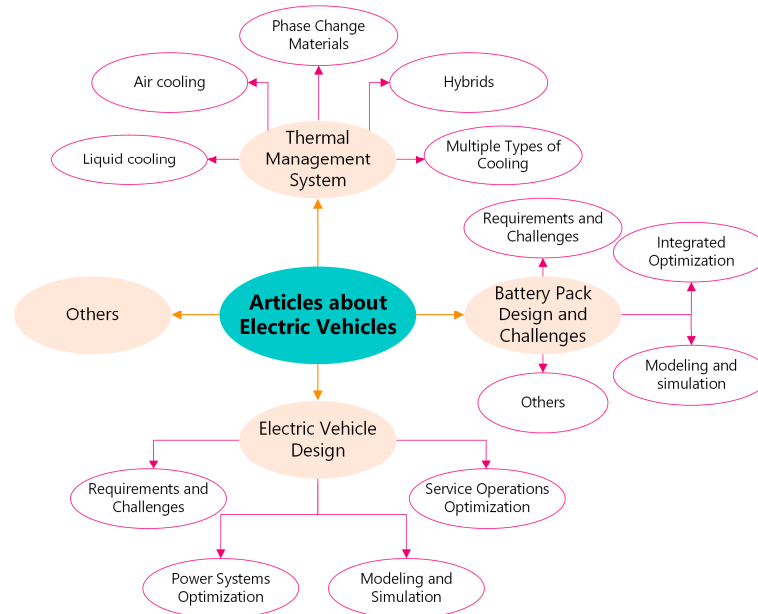


Figure 5. Subareas of electric vehicle articles.

3.2.1. Liquid Cooling for BTMSs

Liquid cooling offers opportunities and significant challenges in the thermal management of electric vehicle battery packs. One of the opportunities is high cooling efficiency, which maintains optimal temperatures during intense charging and discharging. This also enables a more even distribution within the pack, thereby extending cell life. Another critical aspect of liquid cooling is its flexibility, as it can be integrated with other thermal management systems and even incorporate some control.

The primary challenge of liquid cooling systems is their complexity, as they necessitate meticulous designs to prevent leakage and ensure reliable operation. Implementation costs are higher due to the need for additional components such as pumps, heat exchangers, and piping. These systems also require regular maintenance to prevent issues like corrosion and sediment buildup. Furthermore, the use of refrigerants raises concerns, underscoring the need to develop more environmentally friendly alternatives.

Table 2 provides a comprehensive overview of various liquid cooling strategies used in battery thermal management systems, detailing key parameters such as cold plate configurations, battery types, inlet temperatures, flow rates, and heat transfer equations. The table systematically categorizes studies based on different cooling approaches, including configurations where cooling occurs on the bottom, side walls, or radiator-based setups.

Table 2. Summary of research on liquid cooling.

Ref.	Strategies of Cold Plate	Battery Used	Inlet Temperature	Flow Rate	Heat or Temperature Equation	Objectives
[1]	At the bottom, on two side walls, and at the bottom and on two side walls	Lithium iron phosphate (LiFePO ₄) of 50 Ah	(15, 20, 25) °C	(1, 2, 3) L/min	$Q = \frac{1}{V_b} \left[I^2 R_j + IT \frac{\partial U_{ocp}}{\partial T} \right]$	Min T_{max} and T_{diff}
[2]	Two side walls	Lithium-ion	20 °C	0.06266 kg/s	$T_{avg} = \frac{\sum_{i=1}^N T_i}{N}$	Min ΔT and T_{max}

Table 2. Cont.

Ref.	Strategies of Cold Plate	Battery Used	Inlet Temperature	Flow Rate	Heat or Temperature Equation	Objectives
[3]	Top and bottom	LiFePO ₄	20 °C	50 kg h ^{−1}	$(\rho C_p)_{sink} \frac{\partial T}{\partial t} = k_{sink} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right)$	Min T_{max}
[4]	Top and bottom	NMC Li-ion	298.15 K	Velocity = 0.1 m/s with a viscosity = 1.003×10^{-3}	$Q_{gen} = Q_{ir} + Q_{re}$	Min T_{max} and T_{diff}
[5]	Radiator-based and refrigeration-based strategy	NMC and NCA Lithium-ion	25 °C	$\dot{V}p_{min} - \dot{V}p_{max}$ m ³ /s	$\dot{Q}_{cell} = RI^2 + IT \frac{dU}{dt}$	Min $D_{rate,normalized}$
[6]	Bottom with a radiator	Lithium-ion	20 °C	(10, 20, 30 & 40) L/min	$Q_t = hA(T - T_f)$	$\Delta T_{diff_max} < 5$ °C
[7]	Bottom	Lithium-ion	308.15 K	No less than 0.0223 kg/s	$T_{\sigma} = \sqrt{\frac{\sum_N (T_N - T_{ave})^2 A_N}{\sum_N A_N}}$	Min ΔP , T_{max} and T_{σ}
[8]	Top and bottom	Lithium-ion	30 °C	Flow speed = 6 ± 0.8 m/s	—	Min ΔT_{max} and $\Delta T_{max,pack}$

In this area, there is an interesting article [6] on thermal analysis where the heat generation of the cell is given by the following equation:

$$\dot{Q}_{cell} = RI^2 + IT \frac{dU}{dt} \quad (1)$$

$$\dot{Q}_{battery} = N_{cells} \dot{Q}_{cell} \quad (2)$$

The amount of heat produced by the battery pack ($Q_{battery}$) is equal to the sum of the heat produced by each cell's heat (Q_{cell}) multiplied by the number of cells (N_{cells}).

Liquid cooling systems have two primary design types: radiator-based and cooling designs. These two types of designs have different calculation formulations and applications. Radiator-based cooling involves a transport fluid system that removes heat from the battery and dissipates it to the environment through the radiator. This type of system is very effective when the ambient temperature is lower than the battery's target temperature. The main components of the system, as shown in Figure 6, include a fluid pump, radiator, and fan.

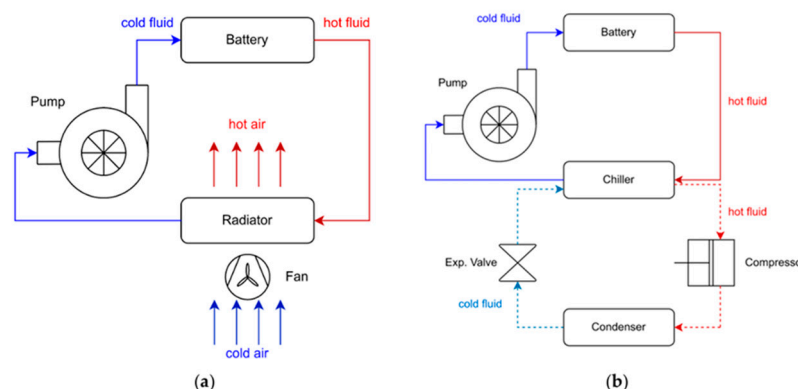


Figure 6. Liquid cooling system types: (a) radiator-based and (b) refrigeration-based system (Reprinted from Ref. [5]).

The heat exchange capacity of a liquid-based cooling system depends on various variables related to the transport fluid, the heat rejection fluid, and the properties of the heat exchanger. The most influential factors include the temperature difference between

the battery pack and the rejection fluid, the surface area and thermal conductivity of the heat exchanger, and the flow rates of the fluids involved.

The power required by the pump and fan is calculated from the determination of the required volumetric flow rate of the fluid, using Equations (3) and (4). The relationship between the volumetric flow rate and the power demanded by this equipment can be represented by means of models based on experimental data or derived formulas.

$$P_{pump} = f(V_{transport}) \quad (3)$$

$$P_{fan} = f(V_{rejection}) \quad (4)$$

The power consumption, in W, of a refrigeration-based system is determined by the sum of the pump's power and the compressor's power.

The case study presented in this article is illustrated in the following graph shown in Figure 7, where the optimization algorithm aims to minimize the objective function, which is the normalized degradation index, using four operating variables, referred to as input variables. As output variables, the coil temperature, as well as the air and coolant flows, were optimized.

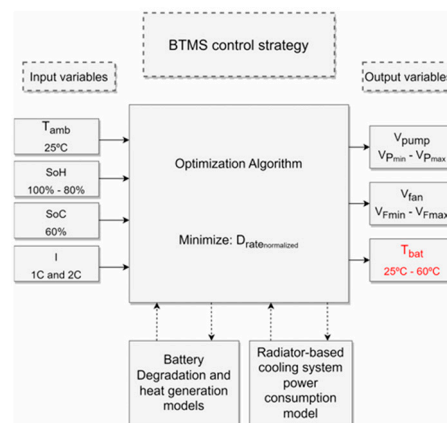


Figure 7. Summary of the case study to which the optimization algorithm was applied (Reprinted from Ref. [5]).

The normalized degradation rate considers both degenerative effects and degradation, as well as heat generation, as presented in Equation (5). By incorporating these two negative phenomena into the same equation, it is now possible to analyze the problem using Equation (5) and optimization algorithms.

$$D_{rate \text{ normalized}} = \frac{D_{rate}}{\frac{Q - (P_{cool} + Q_{battery})_{top}}{Q}} \quad (5)$$

Equation (5) is derived using fundamental physical principles, including the battery's energy conservation and an estimate of the total cycle count based on the specified State of Health (SoH) range and the rate of health decline.

The results showed a 50% improvement in total useful battery cycles in the best-case scenario. The algorithm consistently outperformed the baseline in all conditions tested. The results indicate a reduction in autonomy as the State of Health (SoH) decreases. This phenomenon is attributed to the increase in internal resistance resulting from degradation mechanisms.

3.2.2. Air Cooling for BTMS

Air cooling in the thermal management of electric vehicle batteries presents several opportunities and challenges, according to the articles in this category. Among the opportunities, the simplicity and lower cost of implementation compared to liquid cooling systems stand out, making them attractive for budget-constrained applications. In addition, these systems require less maintenance, as there is no risk of fluid leakage and no need for additional components such as pumps and piping. Air cooling can also be lighter and take up less space, benefiting the design and efficiency of electric vehicles. Furthermore, these systems can be easily integrated with other vehicle components, such as ventilation and air conditioning, improving overall thermal management. However, air-cooled systems face significant challenges, such as lower heat transfer capacity compared to liquid-cooled systems, which can limit their effectiveness under high load and unload conditions. Maintaining an even temperature distribution within the battery pack can also be more difficult, creating problems in very hot climates, and air cooling may not be sufficient. Air cooling is noisier due to its fan-like components, which disadvantages vehicle users.

There are two types of configurations for introducing air to the battery pack: natural convection and forced convection. Below is a brief explanation of both air-cooling configurations. A diagram illustrating the various air-cooling configurations is shown in Figure 8.

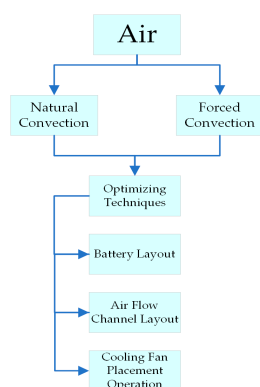


Figure 8. Classification of air cooling with optimization techniques.

Natural Convection

This cooling method occurs when air flows naturally around the battery without mechanical assistance. It relies on temperature differences between the battery and its surroundings, creating air currents due to density variations. Key characteristics include the following:

1. Simple and cost-effective design, as no additional components are needed.
2. Lower efficiency in high-power systems due to limited heat dissipation capacity.
3. Dependence on system orientation and design since air movement follows natural gravitational patterns.

Forced Convection

This approach uses fans or blowers to direct airflow over the battery, enhancing heat dissipation. Its main features are as follows:

1. Greater effectiveness in thermal regulation by actively controlling air movement.
2. Increased energy consumption due to mechanical components.
3. Suitable for high-energy-density systems, where substantial heat removal is crucial.

Table 3 presents a detailed summary of research studies focused on air cooling methodologies, highlighting various design variables, flow rates, heat equations, and optimization

objectives. The table systematically organizes information from multiple sources, specifying key parameters such as cell spacing, cooling channel configurations, and air inlet strategies. The objectives of these studies primarily aim to minimize thermal gradients, pressure drops, and peak temperatures, providing valuable insights into optimizing battery thermal management through air cooling techniques. This table serves as a crucial reference for comparing various approaches to improving cooling efficiency in energy systems.

Table 3. Summary of research on air cooling.

Ref.	Design Variables	Number of Cells	Flow Rate	Heat or Temperature Equation	Objectives
[9]	Cell Spacing, cooling channel size, air supply strategy, others	(8, 88) prismatic, (7, 14, 25, 30, 60, and 66) cylindrical	(0.012) kg/s (0.01, 0.05, 0.1, 0.5) m/s (10.2, 20.4, 30.6, 40.6) m ³ /h	$\dot{Q}_1 = \dot{Q}_{joule} + \dot{Q}_{reaction}$	Min ΔT_{max}
[10]	Air inlet angle, the air outlet angle, and the air flow channel	10 orthogonal	Air flow = 3 m s ⁻¹ Outside air = 5 W m ⁻² k ⁻¹	$\bar{T} = \frac{1}{n} \sum_{k=1}^n T_{maxk}$	Min ΔT and T_{max}
[11]	Cell space combination and channel height	4P128S cylindrical	Flow inlet velocity = 5 m/s	$\dot{Q}_{gen} = \dot{Q}_{irr} + \dot{Q}_{rev}$	Min T_{max} Min T_d Min P s.t. $2 \leq X_i$ $15 \leq X_h$
[12]	Multidisciplinary design optimization (MDO) based on fidelity	10 prismatic	The mass flow rate of the cooling air = (0.001–0.020) kg/s	—	Min T_{max} Min ΔT Min p
[13]	Cell space combination and cooling channel configuration	Cylindrical	Air flow rate = 40.3 m ³ /h Air inlet = 3 m/s	$q = \frac{I^2(R_c + R_p)}{V}$	Min ΔT and T_{max}
[14]	Airflow velocity and cooling channel configuration	8 prismatic	Airflow velocity = 3, 3.5, and 4 m/s	—	Min T_{max} Min ΔT_{max} Min ΔP
[15]	Heat transfer model for cell spacing optimization	45 prismatic	Inlet air flow rates = 0.008, 0.010, 0.012, 0.015 m ³ /s	$\rho_b C_{p,b} \frac{\partial T_b}{\partial t} = \frac{\partial}{\partial x_j} \left[\left(\lambda_b \frac{\partial T_b}{\partial x_j} \right) \right] + \varphi_b$	Min T_{max} Min ΔT_{max}
[16]	Different positions of the inlet vent with and without the flow diverter disc	26,650 cylindrical in a 5P5S configuration	Air speeds = 0.8, 5, and 30 m/s	$Q = \frac{1}{V} \left[I^2 R_i + I T \frac{dU_0}{dT} \right]$	Optimum thermal performance Min T_{max}

3.2.3. Phase Change Materials (PCMs) for BTMSs

Thermal management of electric vehicle batteries using phase change materials presents several opportunities and challenges, as noted in the articles classified in this category. Among the opportunities is the high thermal storage capacity of PCMs, which can absorb and release large amounts of thermal energy during their phase change, allowing batteries to be maintained at optimal temperatures and improving their performance and service life. In addition, PCMs provide constant thermal stability during energy storage and release, which is crucial for applications requiring precise thermal control. Energy efficiency is another advantage, as the ability of PCMs to efficiently store and release thermal energy can contribute to reducing energy consumption and improving the sustainability of electric vehicles. Also, PCMs can be integrated with other thermal management systems, such as liquid or air cooling, to improve overall system efficiency.

Phase change material (PCM) cooling for electric vehicle (EV) battery packs can be divided into three main types: organic, inorganic, and eutectic PCMs. Organic PCMs, such as paraffin-based materials, provide passive thermal regulation, ensuring stability and non-corrosiveness, but they require thermal conductivity enhancers for effective heat dissipation. Inorganic PCMs, including salt hydrates, exhibit higher thermal conductivity and latent heat storage, making them suitable for rapid heat dissipation; however, they may need protective coatings to prevent corrosion. Eutectic PCMs, formed by combining organic and inorganic components, offer precise thermal control with tailored melting points, optimizing battery reliability under varying load conditions. Although their complex formulation often increases costs, these PCMs can enhance battery efficiency, prevent thermal runaway, and contribute to the longevity of EV battery systems. Figure 9 illustrates the classification of PCMs, along with a summary of their advantages and disadvantages.

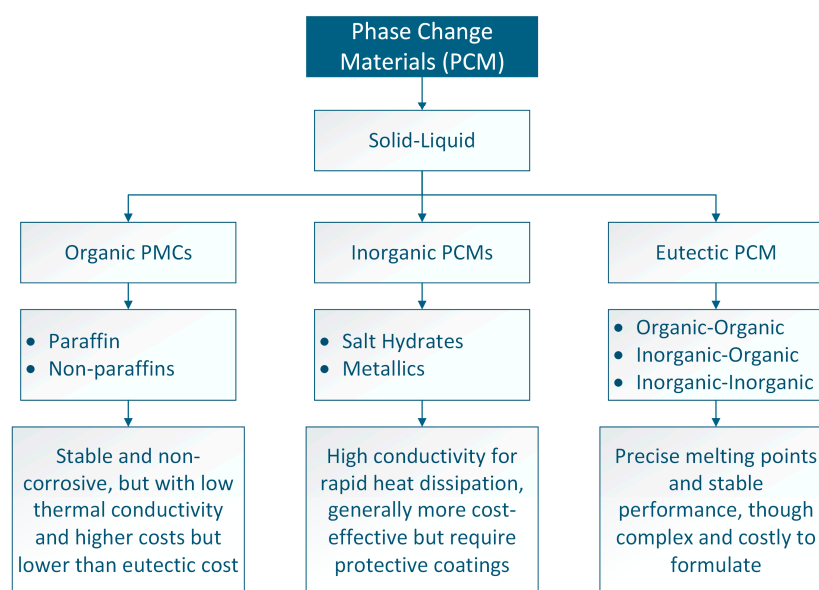


Figure 9. Types of materials used in PCMs and their advantages and disadvantages.

The selection of phase change materials (PCMs) is based on thermal, physical, chemical, kinetic, and economic criteria. It is essential to consider the transition temperature and latent heat for thermal efficiency. Structural properties must allow integration without affecting system stability. Operational safety depends on chemical stability and the absence of toxicity or flammability. In addition, factors such as crystallization and prevention of supercooling ensure reliable performance. Finally, availability and cost-effectiveness determine their viability in industrial and commercial applications. With this evaluation, it is possible to choose PCMs that optimize the thermal management of the systems in which they are implemented. Figure 10 provides a detailed overview of the selection criteria for the materials above.

Sun investigated the thermal management of lithium-ion batteries using a composite PCM-fin structure to improve heat dissipation. While phase change materials (PCMs) effectively absorb heat during phase transitions, their low thermal conductivity limits their efficiency in battery thermal management systems (BTMSs). To address this, the study introduces novel fin structures, including longitudinal fins and cylindrical rings, which enhance heat transfer within PCM-based BTMSs. Experimental and numerical validations confirm that the PCM-fin system significantly outperforms both pure PCM and battery-only systems in controlling temperature rise.

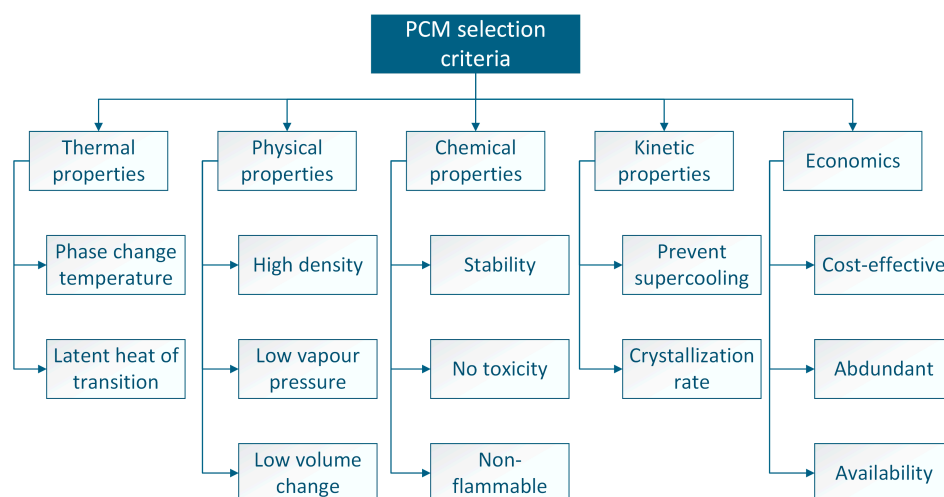


Figure 10. Desirable properties of latent heat storage materials (Reprinted with permission from Ref. [51], Suhanyaa S. Magendran, *Nano-Structures & Nano-Objects*, Published by ELSEVIER, 2019).

The optimized configuration, consisting of eight longitudinal fins and one cylindrical ring positioned at a dimensionless distance of 0.2, delivers the best thermal performance. Additionally, the system effectively manages battery heat generation up to 20 W, demonstrating its suitability for high-power lithium-ion applications. The selected PCM, paraffin wax (OP44E), offers stable thermal properties, including a melting temperature range of 40.6–44.7 °C, a latent heat capacity of 240,800 J/kg, and a thermal conductivity of 0.2 W/(m·K), ensuring efficient cooling performance.

Budiman experimentally investigated the use of phase change materials (PCMs) as thermal barriers in electric vehicle (EV) battery modules to improve thermal management efficiency. Lithium-ion batteries require precise temperature control to maintain optimal performance and prevent overheating, as excessive temperatures above 40–70 °C can degrade battery life and efficiency. The study explores the integration of paraffin-based PCMs into carbon fiber tubes strategically placed within the battery module to minimize non-uniform temperature distribution. Different configurations are tested, including PCMs as an airflow barrier and PCMs as a heat dissipation buffer, revealing that proper PCM placement significantly reduces temperature variations within the module. Experimental results indicate that optimal PCM distribution improves cooling performance, enhances uniform temperature profiles, and extends discharge time, potentially increasing the operational range of EV batteries. The findings suggest that PCM-based BTMSs can complement air-cooling systems in battery modules, offering a practical and lightweight alternative for thermal regulation.

However, PCM-based systems also face significant challenges. The upfront costs of PCMs can be high, especially for organic materials such as kerosene and fatty acids, which can limit their adoption in low-budget applications. Additionally, some PCMs can experience long-term stability issues, such as corrosion in the case of hydrated salts, which can impact their performance and durability. Inorganic PCMs can face subcooling problems, affecting their ability to release thermal energy efficiently. Finally, integrating PCMs into battery thermal management systems requires careful design and innovative solutions to ensure their effectiveness and reliability.

3.2.4. Hybrid Cooling for BTMSs

Hybrid cooling in the thermal management of electric vehicle batteries presents several significant opportunities. First, these systems, which combine liquid and air-cooling methods, offer high heat transfer capacity. This enables the maintenance of batteries at

optimal temperatures, even under intense charging and discharging conditions, thereby improving battery performance and service life. Additionally, the combination of different cooling methods allows for the design of more flexible and adaptable systems to various battery and vehicle configurations, facilitating the customization of thermal management solutions to meet specific needs. The integration of multiple cooling methods can also optimize energy use, reduce power consumption, and improve overall system efficiency. Furthermore, hybrid systems can provide more effective thermal management, reducing the risk of overheating and enhancing battery safety. The research and development of hybrid cooling systems drive technological innovation, opening up new opportunities to strengthen battery thermal management and advance electric vehicle technology.

Saeedipour presents a hybrid battery thermal management system (BTMS) that integrates the phase change material (PCM) aluminum foam and forced-air cooling to improve the thermal performance of cylindrical lithium-ion battery modules. The study addresses the limitations of the low thermal conductivity of PCMs by incorporating aluminum foam, which enhances heat dissipation efficiency. Computational fluid dynamics (CFD) simulations demonstrate that the hybrid system significantly reduces the maximum temperature to 308.1 K and improves temperature uniformity, lowering the temperature difference to 2.0 K. The optimal PCM thickness of 3 mm combined with a 2 m/s airflow and 25 mm cell spacing provides the best cooling results. The PCM used is RT27 paraffin, enhanced with aluminum foam for improved thermal properties, including a thermal conductivity of 4.49 W/(m·K), a latent heat capacity of 179 kJ/kg, and a melting temperature of 300.15 K. The findings indicate that integrating PCM with aluminum foam and forced-air cooling effectively ensures battery temperature stability, uniformity, and safety, making it a promising approach for high-power lithium-ion battery applications.

Wei presents the development and experimental analysis of a hybrid cooling concept for electric vehicle (EV) battery packs, aiming to enhance cooling efficiency and temperature uniformity. The study integrates conductive, convective, and evaporative phase change cooling effects without requiring additional power input beyond conventional air cooling. The proposed system utilizes hydrophilic fiber channels driven by capillary action to transport water coolant, enabling latent heat absorption through evaporation to regulate battery temperature. Additionally, recycled air conditioning (A/C) condensate is used as a sustainable coolant, eliminating extra weight while maintaining environmental friendliness. Experimental results demonstrate that the hybrid cooling system improves cooling efficiency by 70% and temperature uniformity by 72.4% compared to a baseline with no cooling. Moreover, it outperforms traditional air cooling by 20% in efficiency and 56% in uniformity, ensuring reliable thermal management for lithium-ion battery packs in demanding EV applications. The findings indicate that this hybrid approach can effectively prevent thermal runaway and extend battery lifespan, providing a promising solution for next-generation electric vehicle thermal management systems.

Hybrid cooling systems also face several challenges, with system complexity being one of the primary concerns. These systems require careful design to ensure reliable operation. Integrating multiple cooling methods can increase system complexity and implementation costs. Furthermore, hybrid cooling systems can be more expensive to implement compared to other cooling methods due to the need for additional components, such as pumps, heat exchangers, and piping. These systems also require regular maintenance to ensure efficiency and prevent problems, including corrosion and sediment buildup, which can be more complicated and costly due to the system's complexity. Additionally, certain refrigerants used in hybrid systems can have a negative environmental impact, making it crucial to develop and use more environmentally friendly refrigerants to minimize ecological impact. Finally, integrating different refrigeration methods in a single system

can present significant technical challenges, requiring innovative design and advanced technical solutions to ensure system effectiveness and reliability.

In conclusion, Table 4 presents the advantages and disadvantages of each type of refrigeration, along with the specific applications or conditions in which their use is most recommended.

Table 4. Comparative summary of the advantages, disadvantages, and best performances of the different BTMSs.

Cooling Type	Advantages	Disadvantages	Recommended Application
Air	Low cost, easy implementation, and maintenance. It does not require additional components.	Low heat dissipation capacity due to low air conductivity. It can generate temperature gradients inside the battery pack.	Low-power applications in small electric vehicles or hybrids, where simplicity and cost are a priority.
Liquid	Efficient heat dissipation, uniform distribution, and precise temperature control.	Increased complexity and cost due to the need for pumps, piping, and radiators. Risk of leakage.	Essential for high-performance EVs and large-capacity batteries needing thermal stability.
PCM	Ensures stable temperature and safety without extra energy.	Low long-term heat dissipation and material degradation risk.	Ensures stable thermal control in moderate cycling systems without extra power.
Hybrid	Enhances efficiency and stability by merging previous methods.	Increased design and implementation complexity, high costs, and need for precise integration.	Essential for precise thermal control in high-performance EVs and dense batteries.

3.3. Battery Pack Design and Challenges

Battery pack design involves optimizing various factors, including capacity, safety, cost, and lifespan. The studies reviewed explore different approaches and methodologies for enhancing battery pack design. Table 5 provides a classification of articles related to battery pack design and its associated challenges.

Table 5. Classification of the battery pack and challenges.

Area	Ref.	Highlights of the Papers
Requirements and Challenges	[52–63]	Reduce costs, improve performance (drive range), fast charging, and safety.
Integrated Optimization	[64–78]	Optimization framework for battery sizing. Passive cell balancing and sensitivity-based model predictive control (SMPC) approach for optimal and fast charging.
Modeling and Simulation	[79–84]	A dynamic battery energy model optimizes battery pack sizing based on vehicle energy consumption, considering mass, specific energy, and range.
Others	Marketing [85]	Commercialization of lithium battery technologies, obstacles in battery development, including cost reduction, scalability, and longevity, required for widespread EV adoption.
	Design [86,87]	Design a battery pack for a racing application. Clustering-based approach using k-means and support vector clustering (SVC) algorithms.
	Balancing [88–90]	Constant and pulsed active balancing current patterns on the aging of LIBs.

The articles in this category review the requirements and challenges of batteries for electric vehicles. For example, Deng [52] highlights the importance of improving energy density, safety, and lifetime and reducing battery costs. Ahmad [53] discusses the opportunities and challenges of battery swapping stations, highlighting the need for a standardized and cost-effective infrastructure. Omariba [54] reviews cell balancing

methodologies, emphasizing the improvement of efficiency and battery lifetime, although they face challenges in implementation and additional costs.

Benabdelaziz introduces a dynamic energy model for battery systems designed for integration into electric vehicle simulations, emphasizing the need for simulation tools to predict battery performance and minimize physical testing costs. The study categorizes battery models into electrochemical, mathematical, and electrical types, highlighting their advantages and limitations. The authors adopt an energy flow approach using bond graph methodology and MATLAB/Simulink, enabling thermal and electrical analysis of battery behavior under different conditions. The proposed model incorporates electrolyte temperature variations and state-of-charge parameters, offering an accurate depiction of battery performance. Figure 11 shows the block diagram of the battery model, which facilitates an understanding of the energy behavior. In this article, the authors use the graphical linkage method, which enables them to focus on the elements that govern the links between components, thereby avoiding the need to consider mathematical equations.

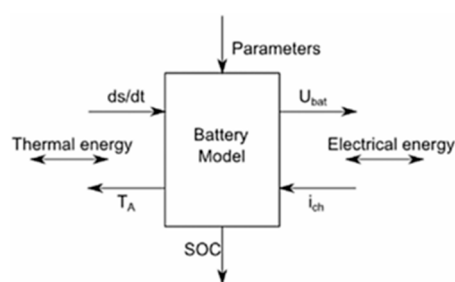


Figure 11. Block diagram of the battery model (Reprinted with permission from Ref. [79], Kawtar Benabdelaziz, *International Journal of Hydrogen Energy*, Published by ELSEVIER, 2017).

Validation is achieved through comparison with previous studies and experimental data, demonstrating the effectiveness of this approach in estimating battery characteristics and optimizing vehicle systems. The findings reveal the impact of ambient temperature and charge/discharge cycles on battery performance, stressing the importance of thermal management for long-term efficiency. The paper concludes that the developed model is a valuable tool for engineers and researchers seeking to improve battery integration within electric vehicles, facilitating more comprehensive system analysis and management strategies. Future work includes expanding the model to incorporate mechanical and motor components in vehicle simulations, ensuring more thorough analysis and optimization capabilities.

The primary challenges in developing and optimizing batteries for electric vehicles include cost reduction, which remains a significant obstacle to the commercialization of advanced battery technologies. Complexity in designing and implementing thermal and energy management systems also presents a substantial hurdle, necessitating innovative and efficient solutions. Furthermore, enhancing the safety and sustainability of batteries is crucial, particularly in terms of recycling and minimizing environmental impact. Another critical challenge is the need for adequate and standardized infrastructure, particularly for battery swapping stations and fast charging systems. Finally, improving the efficiency and lifetime of batteries is a key objective, but it faces significant technical challenges that must be addressed to achieve optimal and long-lasting performance.

3.4. Electric Vehicle Design

Electric vehicle design encompasses various aspects, from modeling and simulation to charging station planning and powertrain optimization. As shown in Table 6 a classification of the subareas important for electric vehicle design is presented.

Table 6. Classification of the subareas of the electric vehicle design.

Areas	References
Requirements and Challenges	[91]
Power System Optimization	[92,93]
Modeling and Simulation	[94,95]
Service Operations Optimization	[96]

Sanguesa provides a comprehensive review of electric vehicles (EVs), focusing on technological advancements, battery innovations, charging methods, and emerging challenges. It examines the current global market situation, highlighting growth trends, subsidies, and government policies promoting EV adoption. A detailed analysis of battery technologies is presented, ranging from lead-acid to lithium-ion, alongside next-generation alternatives like graphene and sodium-air batteries. The study also explores charging standards and infrastructure, comparing global approaches, while discussing fast-charging and wireless charging advancements. Furthermore, it investigates battery management systems (BMSs), thermal regulation, and power electronics, emphasizing optimization strategies for efficiency and safety. The review addresses critical challenges such as charging infrastructure limitations, battery recycling concerns, and environmental impact while proposing future research directions, including AI-driven energy management, smart grid integration, and sustainable mobility solutions. Overall, the study offers insights into the evolving EV landscape, identifying key technological innovations and market trends shaping the future of electromobility.

Jung discusses the advantages and challenges of transitioning from 400 V to 800 V systems in battery electric passenger vehicles, highlighting the potential benefits, including reduced battery charging time, weight savings due to lower current requirements, and improved vehicle efficiency. By doubling the voltage, the current required for the same power level is halved, enabling smaller wiring harnesses, reduced copper losses, and increased thermal efficiency. However, challenges such as the limited availability of automotive-grade 800 V components, increased development costs, and modifications to existing infrastructure must be addressed for widespread adoption. The study also examines battery pack configurations, showing that while voltage increases, the number of cells remains the same, requiring redesigns of power inverters, chargers, and thermal management systems. Additionally, it explores fast charging technology, illustrating how an 800 V system can reduce charge times from 29 min to less than 15 min, making electric vehicles (EVs) more practical for long-distance travel. Future developments in battery technology, charging infrastructure, and regulatory standards will play a crucial role in the successful implementation of 800 V systems in mainstream electric mobility.

Tran presents the design and optimization of a hybrid electric vehicle (HEV) powertrain, evaluating various powertrain components and configurations to enhance performance. The study analyzes series, parallel, and series-parallel hybrid architectures, highlighting their advantages in fuel economy and emissions reduction. Using MATLAB/Simulink simulations, five different designs were tested to meet acceleration, braking, range, and fuel efficiency criteria set by the EcoCAR Mobility Challenge. The final optimized powertrain consists of a P4 hybrid configuration, incorporating a 2.5 L GM engine, a 150 kW electric motor from AAM, and a 133 kW battery pack from HDS, ensuring superior fuel economy and reduced emissions.

Alegre presents the modeling and optimization of electric and parallel-hybrid electric vehicles (HEVs) using MATLAB/Simulink and the planning of charging stations using a geographic information system (GIS) and genetic algorithms. The study examines key vehicle parameters such as battery type, aerodynamics, energy consumption, and driving cycles to analyze EV autonomy under different conditions. Additionally, it proposes an optimized approach for charging station placement, minimizing installation costs while ensuring efficient geographic distribution to enhance service reliability. The optimization method applies genetic algorithms to determine ideal locations based on charging station power, demand forecasting, and urban constraints. Simulation results validate the effectiveness of the proposed methods, showing improvements in battery performance, vehicle autonomy, and charging infrastructure efficiency, contributing to the sustainable growth of electric mobility.

Reviewing the requirements and challenges category, the articles emphasize the importance of improving energy density, safety, and lifetime while reducing costs. They also discuss the opportunities and challenges of battery swapping stations, highlighting the need for standardized and cost-effective infrastructure. These studies provide a comprehensive view of the critical issues that must be addressed to advance electric vehicle technology and make it more accessible and efficient.

The hybrid powertrain design is also analyzed to optimize performance by considering various components and configurations. These studies are crucial for developing more powerful and efficient electric vehicles that can compete with internal combustion vehicles in terms of performance and range.

In modeling and simulation, tools such as Matlab/Simulink are used to model and analyze vehicle performance. Energy consumption is also studied, and charging stations are planned using geographic information systems and genetic algorithms. These studies are fundamental in predicting the behavior of electric vehicles under different conditions and optimizing their design and operation.

In the optimization of service operations, the parameters and costs associated with battery electric vehicles are reviewed. Optimization models for service operations are also analyzed, highlighting the importance of improving operational efficiency. These studies are essential for developing strategies to reduce operating costs and improve the economic viability of electric vehicles.

3.5. Other

This category includes articles that do not fit directly into the previous categories but are relevant to electric vehicles and batteries. Delmas's article [97] provides an overview of the research conducted on sodium batteries over the last 50 years. Pózna [98] proposes a design of experiments to estimate battery aging. In the paper by Momen [99], the primary focus of the research was to develop an efficient and high-performance propulsion system for the Chevrolet Bolt BEV, enhancing the vehicle's range while maintaining excellent acceleration performance. They focused on optimizing the propulsion system components, including the traction motor, power electronics, and energy storage, to significantly improve energy, power, torque, and efficiency.

4. Discussion

The results obtained in this article highlight the opportunities and challenges associated with the thermal management of battery packs for electric vehicles, the development of electric vehicles, and the design of battery packs. Thanks to this study, the use of phase change materials (PCMs) and hybrid systems offers some significant benefits in improving battery performance and lifetime. However, the main challenge is focused on technical and

economic issues, as they have high initial costs and long-term stability problems, such as corrosion due to hydrated salts, which hinder their effective implementation. One of the main advantages of PCMs is their high thermal storage capacity, allowing them to stabilize the operating temperature of the batteries, improve the energy efficiency of the battery pack, and ensure safety. The need for standardized infrastructures, particularly for battery swapping stations and fast charging systems, emerges as a key challenge. Additionally, cell balancing techniques present opportunities to improve battery efficiency and durability, although their implementation implies additional costs and technical challenges.

Further research may explore the integration of alternative battery chemistries, such as solid-state and sodium-ion technologies, to address concerns related to energy density and lifecycle sustainability. Future work could leverage smart grid connectivity to develop dynamic charging station allocation methods based on real-time demand forecasting and urban traffic analytics in the context of charging infrastructure. Additionally, multi-objective optimization algorithms could maximize charging efficiency, minimize installation costs, and ensure equitable geographic distribution of charging stations. The study builds upon prior investigations into EV energy consumption modeling, GIS-based urban planning for charging networks, and genetic algorithms for system optimization, serving as a foundation for advancing next-generation electric mobility solutions.

5. Conclusions

Thermal management is a key factor in the design and operation of electric vehicles, as it directly influences the efficiency, safety, and durability of batteries and other electronic components. Over time, several researchers have explored different approaches and technologies to optimize thermal management in this type of vehicle. The relevance of proper thermal management lies in keeping batteries and electronic systems within an optimal temperature range, avoiding overheating, and improving energy efficiency. To this end, multiple solutions have been developed, such as liquid cooling systems, phase change materials (PCMs), and strategies based on artificial intelligence to optimize heat dissipation and distribution.

However, some of the main challenges in this area include the complexity of the systems, the associated costs, and their integration with other elements of the vehicle. To address these challenges, the specialized literature has proposed the use of advanced materials, thermal control algorithms, and integrated design approaches that seek to improve efficiency and overcome current limitations. Future lines of research focus on improving the accuracy of thermal models, developing more efficient and sustainable thermal management systems, and exploring emerging technologies such as immersion cooling and nanotechnology. In this context, cooperation between academic institutions and industry will be key to driving advances and generating innovative solutions applicable to future electric vehicles.

Future research could explore alternative battery chemistries like solid-state and sodium-ion technologies to enhance energy density and sustainability. Smart grid integration may enable dynamic charging station allocation based on real-time demand and urban traffic analysis. Additionally, multi-objective optimization could improve efficiency, reduce costs, and ensure equitable distribution, advancing next-generation electric mobility.

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